MANDATORY SPARES LIST (ELECTRICAL)

EPCC-03 PACKAGE FOR RESIDUE UPGRADATION FACILITY UNIT (RUF)

TENDER NO. B016-504-84-41-PM-T-9509

PROJECT : VISAKH REFINERY MODERNISATION PROJECT

UNIT : 504

OWNER : HPCL VISAKHAPATNAM

PMC : EIL

JOB NO. : B016

Rev. No	Date	Purpose	Prepared by	Checked by	Approved by
A	22.05.2018	ISSUED FOR TENDER	NNB	AS	PG

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LIST OF MANDATORY SPARES: ELECTRICAL

SR.NO.	PART DESCRIPTION	QUANTITY
1.	**GAS INSULATED SWITCHGEAR**	**one set of spare for each switchgear**
1.1	Portable gas filling equipment/SF6 gas cart	-
1.2	Handle for disconnector switch drive	1 Nos.
1.3	Handle for earthing switch drive.	1 Nos.
1.4	Pre selection key for three position switch.	1 Nos.
1.5	Power cable termination kit	1 Sets
1.6	Tripping coil (Per Switchgear)	1 Nos.
1.7	Closing coil (Per Switchgear)	1 Nos.
1.8	Capacitive type voltage detectors	1 Set
1.9	Control fuses/MCB	10 Nos. each rating and type
1.10	Density monitoring device	2 Nos. each type
1.11	Pressure Gauge	2 Nos. each type
1.12	Indication lamp covers	5 Nos. of each colour
1.13	Indication lamps	20% or 3 nos. (min) whichever is more
1.14	Network switches	1 No. of each type
2.	**HV SWITCHGEAR (33KV/11KV/6.6KV)**	**one set of spare for each switchgear**
2.1	Closing coil	1 Nos. each rating/ type
2.2	Shunt trip coil	1 Nos. each rating/ type
2.3	control fuses/MCB	10 Nos. each rating and type
2.4	Indication lamp covers	5 Nos. of each colour
2.5	Indication lamps	20% or 3 nos. (min) whichever is more
3.	**MV SWITCHBOARD-415V (PCC/ EPMCC/ MCC/ ASB/ LDB)**	**one set of spare for each switchgear**
3.1	Closing coil	1 Nos. each rating/ type
3.2	Shunt trip coil	1 Nos. each rating/ type
3.3	control fuses(all type & rating)	10 Nos. each rating and type
3.4	Indication lamp covers	5 Nos. of each colour
3.5	Indication lamps	20% or 3 nos. (min) whichever is more
4.	**POWER TRANSFORMER**	**one set of spare for each power transformer**
4.1	Complete set of Gaskets	1 Set
4.2	Sealing/ Gauge glass of conservator	2 Nos of each rating & type.
4.3	Control fuses/MCB for MB cubicles	20% of each rating or 1 No. (min) of each rating, whichever is more.

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5.	**DISTRIBUTION TRANSFORMER**	**one set of spare for each distribution transformer**
5.1	Complete set of Gaskets	1 Set
5.2	Sealing/ Gauge glass of conservator	2 Nos of each rating & type.
5.3	Control fuses/MCB for MB cubicles	20% of each rating or 1 No. (min) of each rating, whichever is more.
6.	**VARIABLE FREQUENCY DRIVE/SOFT STARTER**	**one set of spare for each VFD**
6.1	Control cards	1no. of each type
6.2	Power supply cards	1no. of each type and rating
6.3	Power fuses	20 % or one no (min) of each rating & type, whichever is more
6.4	Control fuses/ MCB	20 % or one no (min) of each rating & type, whichever is more
6.5	Transistor/ IGBT	1no. of each type and rating
6.6	Indication lamp covers	5 Nos. of each colour
6.7	Indication lamps	20% or 3 nos. (min) whichever is more
6.8	Blocker Diode	2 Nos. of each rating & type
7.	**SUBSTATION AUTOMATION (SAS) /DATA CONCENTRATOR PANEL/HMI**	**one set of spare for each SAS/ HMI/ DATA CONCENTRATOR**
7.1	All cards such as input & output cards, power supply cards, processor cards etc.	1 Nos. each type
7.2	Ethernet switches	1 Nos. each type
7.3	Control fuses/ MCB	10 Nos. each rating and type
8.	**Relays For Switchboard/ Relay Control Panel**	One set of spare for each switchboard/ Relay control panel
8.1	Protection Relays	1 No. of each type
8.2	Auxiliary Relays	1 No. of each type
9.	**UPS (one set of spare for each UPS System)**	
9.1	Power thyristors/ transistors/ IGBT/ IGCT	1 Nos. each rating and type
9.2	Control cards	1no. of each type
9.3	Power supply cards	1no. of each type and rating
9.4	Operating Console card	1 No. of each type
9.5	All fuses/ MCB	10 Nos. of each rating &

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		type
9.6	Indication lamp covers	2 Nos. of each colour
9.7	Indication lamps	10% or 3 nos. (min) whichever is more
9.8	Blocker Diode	2 Nos. of each rating & type
10.	**DC SYSTEM**	**One set of spares for each DC System**
10.1	Power thyristors/ transistors/ IGBT/ IGCT	1 Nos. each rating and type
10.2	All cards such as input & output cards, power supply cards, processor cards etc.	1no. of each type and rating
10.3	All fuses / MCB	10 Nos. of each rating & type
10.4	Indication lamp covers	2 Nos. of each colour
10.5	Indication lamps	10% or 3 nos. (min) whichever is more
10.6	Blocker Diode	2 Nos. of each rating & type
10.7	Fuse Links	10 Nos. of each rating & type
11.	**THYRISTOR CONTROL PANELS FOR HEATERS)(IF APPLICABLE)**	
11.1	Rectifier Control Module (Control Card Fully Assembled)	1no. of each type
11.2	Power Supply card	1no.of each type
11.3	Power fuses	20 % or one no (min) of each rating & type, whichever is more
11.4	Control fuses/ MCB	10 Nos. of each rating & type
11.5	Contactors	10% of each type OR 1 No. (min) of each type, whichever is more.
11.6	Indication lamp covers	2 Nos. of each colour
11.7	Indication lamps	20% or 3 nos. (min) whichever is more
11.8	Blocker Diode	2 Nos. of each rating & type
12.	**SYNCHRONOUS MOTORS (IF APPLICABLE)**	**One set of spare for each type and rating**
12.1	All fuses/ MCB	10 Nos. of each rating & type
12.2	Bearing set (DE & NDE both) (as applicable)	1 no. Each type
12.3	Control cards	10 Nos. of each type
13.	**HV INDUCTION MOTORS**	**One set of spare for each type and rating**
13.1	Bearing set (DE & NDE both).	1 Set
13.2	Terminal studs/ bushing assembly	One set each

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14.	**MV INDUCTION MOTORS (37 KW and above)**	**One set of spare for each type and rating**
14.1	Bearing Set (DE &NDE)	1 Set
14.2	Terminal studs / bushing assembly	1 set for each type
15.	**TELEPHONE EXCHANGE**	
15.1	All line cards, processing cards, power supply cards	1 no of each type
16.	**FIRE ALARM SYSTEM**	
16.1	All cards	1 No. of each type
16.2	All fuses / MCB	10 Nos. of each rating & type
16.3	Terminal blocks	20 Nos.
16.4	Fire detectors	5% of total installed capacity of each type
16.5	Glass for Break glass Boxes/ Manual call point	10% of each type,
17.	**PLANT COMMUNICATION SYSTEM**	
17.1	All cards	1 No. of each type
17.2	Power fuses	10 Nos. of each rating & type
17.3	Field Call Stations	10 % of each type
18.	**ECS**	
18.1	Interposing Relays (As applicable)	5 Nos. of each type
18.2	Power Supply & control cards	1 No. of each type
18.3	Transducers	20 % of estimated quantity of each type or one no (min) of each make & type, whichever is more
18.4	Functional generator cum counter	1 No.
18.5	4-20mA signal injection set	1 No.
18.6	Isolation Transformer	1 No.

Notes:-

1. **Mandatory spares list is applicable only for the items supplied by the Contractor.**
2. The word ‘TYPE’ means the Make, Model no., Type, Range, Size/ Length, Rating, Material as applicable.
3. Wherever % age is identified, Contractor shall supply next rounded figure.
4. The terminology used under ‘Part Description’ is the commonly used name of the part and may vary from manufacturer to manufacturer.
5. Mandatory spares as indicated above do not cover commissioning spares.
6. Mandatory spares as indicated above do not cover two year O&M spares.
7. Mandatory spares shall be applicable for electrical items of motors / sub-packages as per mandatory spares philosophy specified elsewhere in the bid document.
8. For Isolation breaker panel (GIS or AIS) one set of “Tripping Coil (1 No.), Closing coil (1 No.) and Control fuse/MCB (10 Nos. of each rating and type) “shall be considered as mandatory spares for each Isolation breaker panel (GIS or AIS).
9. For Isolation breaker panel (GIS or AIS) one set of “Auxiliary relays (1 no. of each type)” shall be considered as mandatory spares for Relays for Isolation breaker panel (GIS or AIS).

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